Week 4 · Assignment 1

EKF vs UKF in a Nonlinear Range–Bearing Tracking System | Information Filter Derivation

Course: Autonomous Driving / Gaussian Filters II • Student: _____ • Date: _____

Status: Experimental report based on a reconstructed Python implementation, not an exact reproduction of the instructor's distributed program. Results are from three seeds (42–44).

Objective and experimental model

The lecture asks for: (1) EKF/UKF RMSE comparison; (2) removal of bearing wrapping; (3) 10× and 100× initial covariance; (4) UKF alpha sweep; (5) NEES with chi-square confidence limits; and (6) derivation and commutativity of the information-form measurement update. [1]

State: $x = [p_x, p_y, v_x, v_y]^T$. Constant-velocity dynamics use $\Delta t = 0.1$ s for 200 steps, $x_k = A\,x_{k-1} + w_k$. The nonlinear sensor observes $z = [r, b]^T$, with $r = \sqrt{(p_x^2+p_y^2)}$, $b = \text{atan2}(p_y,p_x)$. Measurement standard deviations are 0.5 m and 4°. The simulated target passes approximately 0.5 m above the sensor origin near t = 10 s. Process covariance is diag(0.001, 0.001, 0.002, 0.002).

The EKF uses the measurement Jacobian; the UKF propagates nine sigma points (n = 4). Both process the same simulated measurements for each seed. The reconstructed code includes a Joseph-form EKF covariance update, a circular UKF bearing mean, and an origin guard for the Jacobian; these choices may differ from the lecture's illustrative code.

Q1 · Baseline EKF–UKF comparison

Filter	Mean position RMSE (m)	Sample SD (m)	Mean NEES
EKF	0.1740	0.0172	1.343
UKF	0.1737	0.0169	1.341

Interpretation. The UKF RMSE is lower by 0.0003 m (0.3 mm), an extremely small difference relative to the between-seed variation. Both methods tracked this trajectory similarly; these three trials do not establish a statistically meaningful UKF accuracy advantage. Position RMSE is $\sqrt{[\text{mean}((\hat{p}_x-p_x)^2+(\hat{p}_y-p_y)^2)]}$ across steps, then summarized across seeds.

Q2 · Removing angle wrapping

Metric	Wrapping ON	Wrapping OFF
EKF mean RMSE	0.1740 m	5.5945 m
UKF mean RMSE	0.1737 m	5.5933 m
EKF mean NEES	1.343	994.284
UKF mean NEES	1.341	994.001

A bearing is periodic: −179° and +179° differ by 2°, not 358°. The wrapped residual is $\text{wrap}(\delta) = (\delta + \pi) \bmod (2\pi) - \pi$. Without wrapping, the first raw innovations exceeding π were at step 2 (seed 42: −6.1902 rad), step 3 (seed 43: approximately −6.1989 rad), and step 2 (seed 44: approximately −6.1889 rad). The seed-42 innovation wraps to about +0.0930 rad. These are the first large residuals detected, not independently verified first divergence times. Without normalization, a near-2π discontinuity is mistaken for a huge correction, producing roughly 32× larger position RMSE and severe inconsistency.

Q3 · Initial covariance sensitivity

Initial P_0 multiplier	EKF RMSE (m)	UKF RMSE (m)	EKF NEES	UKF NEES
×1	0.1740	0.1737	1.343	1.341
×10	0.1851	0.1846	1.387	1.383
×100	0.1931	0.1927	1.430	1.430

Multiplying covariance by 100 multiplies standard deviation by 10. Increased prior uncertainty makes a local EKF Jacobian potentially less representative of the full state distribution, the lecture's "second factor" behind linearization error. Here both filters' RMSE rose modestly (about 11% from ×1 to ×100), without observed divergence. Thus the experiment supports sensitivity to initialization but does not demonstrate pronounced EKF failure in this implementation.

Q4 · UKF sigma-point spread α

α	UKF RMSE (m)	Mean NEES	Cholesky failure
0.001	0.1737	1.341	No
0.01	0.1737	1.341	No
0.1	0.1737	1.341	No
0.5	0.1737	1.339	No
1.0	0.1736	1.336	No

The sigma-point scaling is $\lambda = \alpha^2(n + \kappa) - n$, and sigma-point offsets use a matrix square root of $(n + \lambda)P$. Larger α spreads points farther from the mean. Performance changed negligibly across the tested range: the 0.0001 m RMSE improvement at $\alpha = 1.0$ is not evidence of a universally optimal setting. All five runs completed without a Cholesky failure; this does not guarantee stability for other trajectories or seeds.

Q5 · NEES and consistency

For each step, $\text{NEES}_k = (\hat{x}_k - x_k)^T P_k^{-1}(\hat{x}_k - x_k)$. For a consistent four-dimensional state estimate, $E[\text{NEES}] = 4$. The pointwise average over $N = 3$ independent runs is $\text{ANEES}_k = (1/N)\Sigma_i \text{NEES}_{k,i}$. Under the usual Gaussian/calibration assumptions, $N \cdot \text{ANEES}_k$ follows χ^2 with $Nn = 12$ degrees of freedom. Its 95% pointwise interval is $[\chi^2_{12,0.025}/3, \chi^2_{12,0.975}/3] \approx [1.468, 7.779]$.

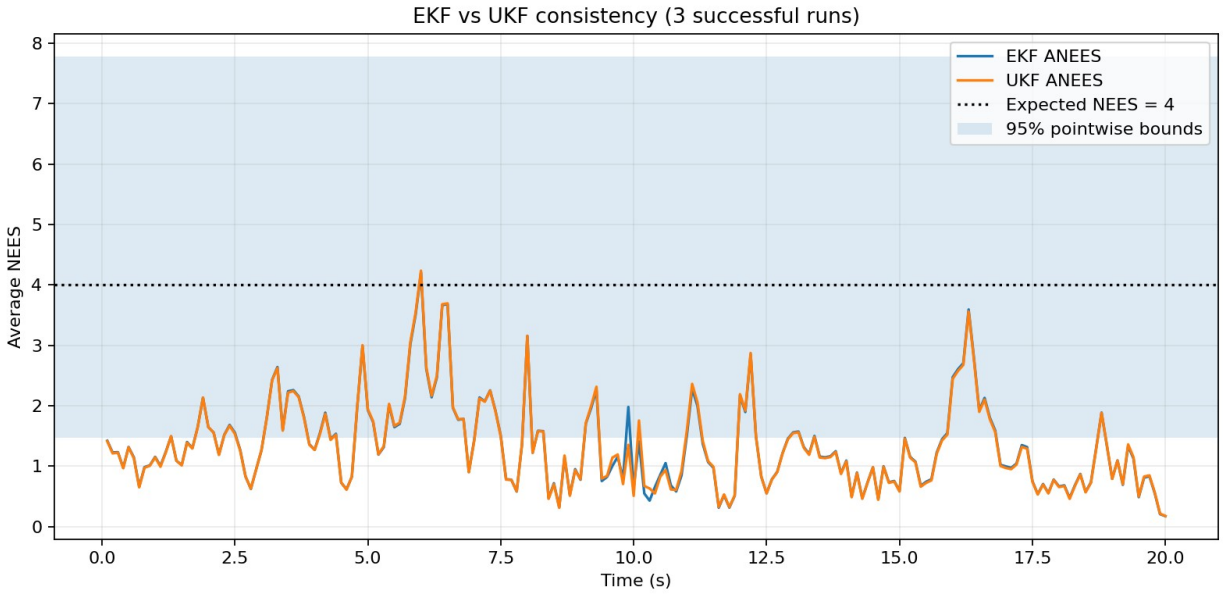


Figure 1. Average NEES by time step for EKF and UKF (three runs); shaded area denotes pointwise 95% bounds.

Both curves nearly overlap, with many steps below the lower bound and occasional peaks near or slightly above the expectation of 4. Overall time-averaged NEES is 1.343 (EKF) and 1.341 (UKF), suggesting possible conservative covariance estimates. The pointwise bounds must not be used as a confidence interval for a time average, and three runs provide weak evidence for formal consistency. Neither filter can be declared more "honest" on these results.

Q6 · Information-form update and measurement-order proof

The lecture defines canonical Gaussian parameters $\Omega = \Sigma^{-1}$ (information matrix) and $\xi = \Sigma^{-1}\mu$ (information vector), with $p(x) \propto \exp(-\frac{1}{2}x^T\Omega x + x^T\xi)$. For a linear observation $z = Hx + v$, $v \sim N(0,Q)$, the Gaussian likelihood contributes $-\frac{1}{2}(z-Hx)^TQ^{-1}(z-Hx)$ to the log posterior. Expanding the quadratic gives $-\frac{1}{2}x^TH^TQ^{-1}Hx + x^TH^TQ^{-1}z$ + terms independent of x. Adding this to the prior canonical exponent proves:

$\Omega^+ = \Omega^- + H^TQ^{-1}H; \quad \xi^+ = \xi^- + H^TQ^{-1}z.$ (linear measurement)

For the lecture's nonlinear h(x), linearize once at the fixed predicted mean $\bar{\mu}$: $h(x) \approx h(\bar{\mu}) + H(x-\bar{\mu})$. This is an affine observation $z \approx Hx + [h(\bar{\mu})-H\bar{\mu}] + v$. Its information-vector increment is therefore $H^TQ^{-1}[z-h(\bar{\mu})+H\bar{\mu}]$, exactly the term in lecture equation (4.7); the information-matrix increment remains $H^TQ^{-1}H$. [1]

Now consider independent observations (z_1,H_1,Q_1) and (z_2,H_2,Q_2). Define $\Delta\Omega_i = H_i^TQ_i^{-1}H_i$ and $\Delta\xi_i = H_i^TQ_i^{-1}z_i$ (or use the fixed-linearization affine form above). Updating 1 then 2 yields $\Omega_{12} = \Omega^- + \Delta\Omega_1 + \Delta\Omega_2$ and $\xi_{12} = \xi^- + \Delta\xi_1 + \Delta\xi_2$. Updating 2 then 1 yields $\Omega_{21} = \Omega^- + \Delta\Omega_2 + \Delta\Omega_1$ and $\xi_{21} = \xi^- + \Delta\xi_2 + \Delta\xi_1$. Since matrix/vector addition is commutative, $\Omega_{12} = \Omega_{21}$ and $\xi_{12} = \xi_{21}$; hence the recovered μ and Σ are identical.

Why is this not obvious for the nonlinear EKF? In a sequential EKF, the second measurement can be linearized around the state already corrected by the first, changing its Jacobian H and predicted measurement h(μ). Reversing the order can therefore change both linearizations and the final approximation. If the same linearization point is held fixed, the independent-measurement information increments do commute; exact linear-Gaussian Kalman updates also yield the same posterior regardless of order.

Conclusion and limitations

In the reconstructed experiment, EKF and UKF showed nearly identical baseline accuracy and consistency. Proper bearing wrapping was essential; increasing initial uncertainty mildly increased errors; changing α had little impact. The NEES analysis suggests conservatism but is limited by three Monte Carlo runs. The information-form measurement update is additive, with order independence for independent observations under a shared linear or fixed-linearized model.

Limitations: Results are not the instructor's original simulator output. Only three seeds were tested, rounded summary statistics obscure tiny differences, and the reconstructed code's numerical safeguards affect behavior. A stronger study would increase the number of independent runs and evaluate sensitivity near the origin.

References and reproducibility

[1] Course lecture: "4 주차 · 가우시안 필터 II — EKF · UKF · 정보 필터," §4.4–4.5 and "연습문제 (과제 1 · 4 주차 제출)," lecture PDF supplied with this coursework.

[2] Reconstructed scripts: ekf_vs_ukf1.py and ekf_vs_ukf_nees.py; output: nees_consistency.csv and nees_consistency.png. Baseline command: py ekf_vs_ukf_nees.py --runs 3 --seed 42 --plot. Other flags: --nowrap, --sigma0 10/100, --alpha 0.001/0.01/0.1/0.5/1.0.